

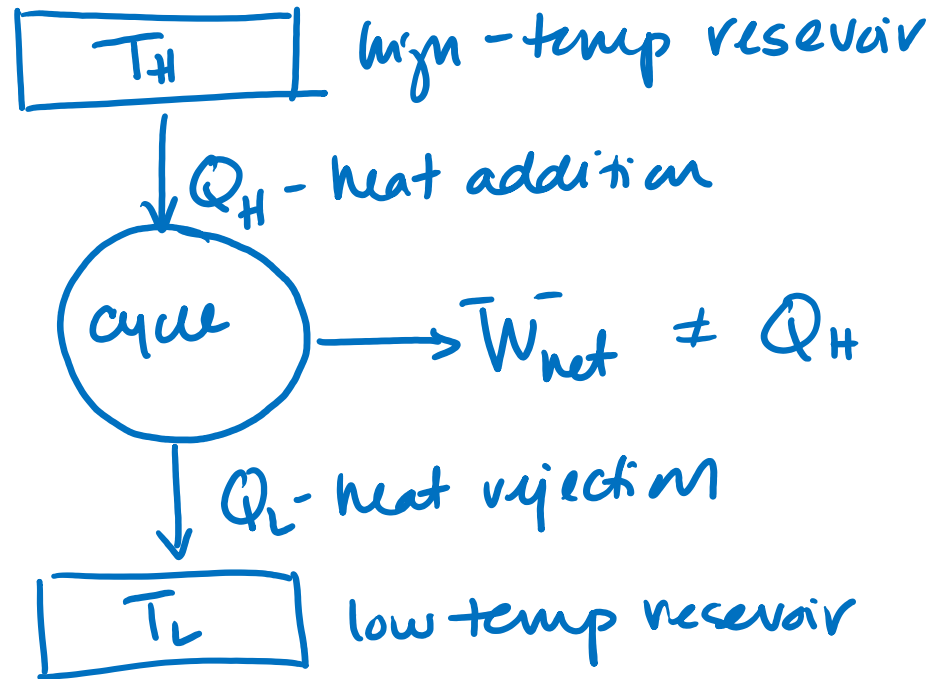


ME 300:
Engineering Thermodynamics

Heat Engines and Thermal Efficiency

Implications of Kelvin Planck

→ cyclically-operating device - heat engine

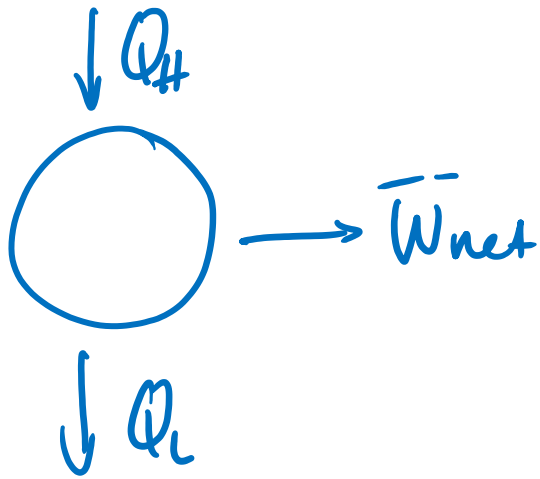


Energy Conservation

→ heat addition: Q_H

net work: $\bar{W}_{net}$

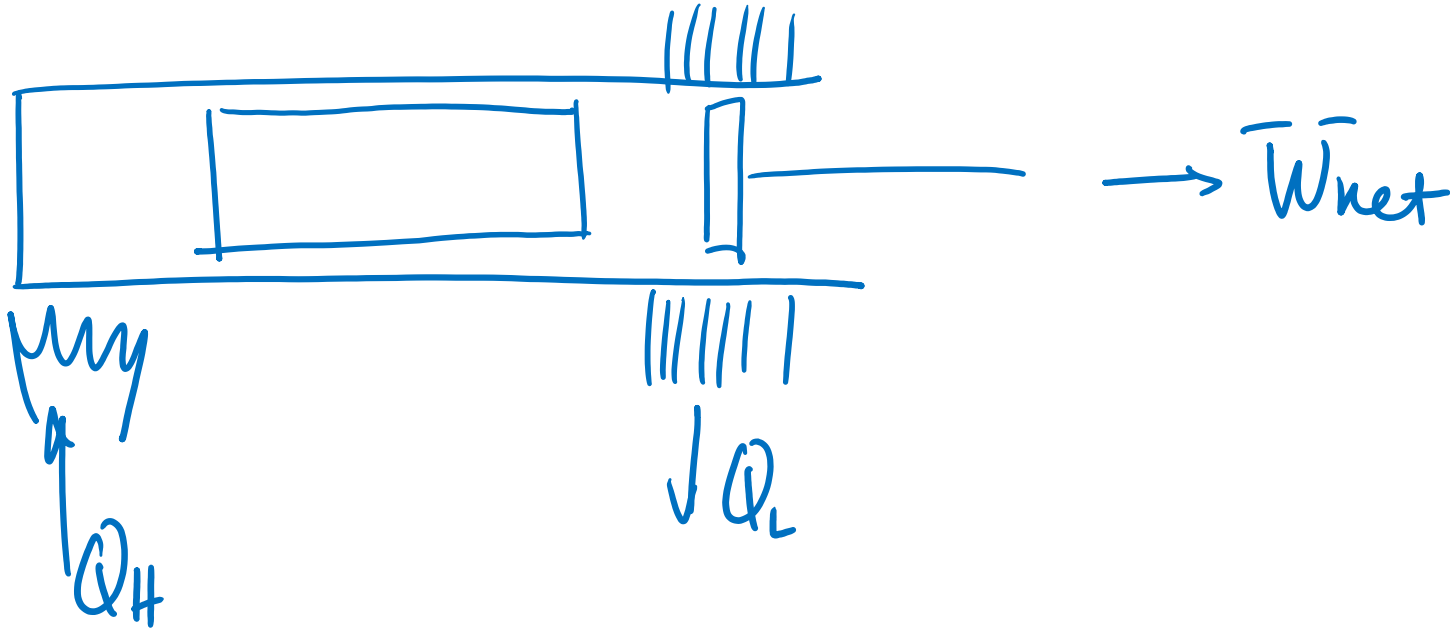
heat rejection: Q_L



$$Q_H = \bar{W}_{net} + Q_L$$

$$\bar{W}_{net} = Q_H - Q_L$$

Example: Stirling Engine



Thermal efficiency

$$\eta_{th} = \frac{\text{net work}}{\text{heat in}} \quad \star \underline{\underline{\text{cycle}}}$$

$$\eta_{th} = \frac{\bar{W}_{net}}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

$$\boxed{\eta_{th} = 1 - \frac{Q_L}{Q_H}}$$



Can the thermal efficiency ever be equal to 1?

A. Yes

B. No